

Aim: Implement Binary Search Tree and its traversals.

CODE:

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;
class Node{
    public:
        int key;
        Node *left = NULL, *right = NULL;
        Node(int x){
            key = x;
            left = right = NULL;
        }
};

void inorder(Node *root){
    if(root!=NULL){
        inorder(root->left);
        cout<<root->key<<" ";
        inorder(root->right);
    }
}

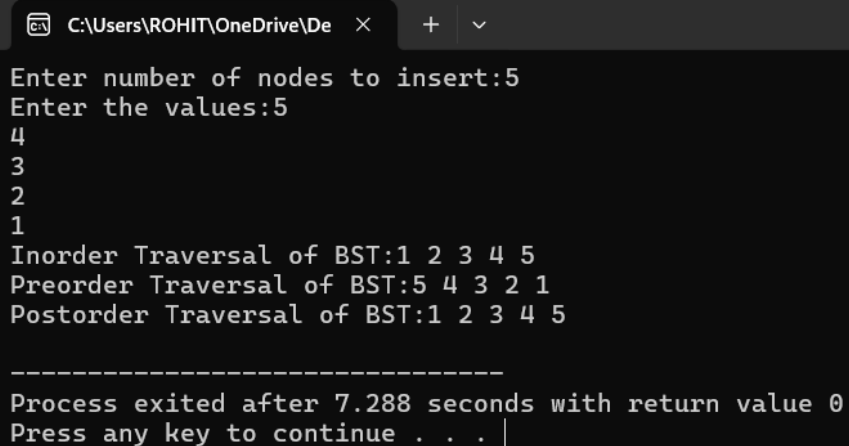
void preorder(Node *node){
    if(node == NULL)
        return;
    cout<<node->key<<" ";
    preorder(node->left);
    preorder(node->right);
}

void postOrder(Node *node){
    if(node == NULL)
        return;
    postOrder(node->left);
    postOrder(node->right);
    cout<<node->key<<" ";
}

Node *insert(Node *node,int key){
    if(node == NULL)
        return new Node(key);
    if(node->key == key)
        return node;
    if(key>node->key)
        node->right = insert(node->right,key);
    else
        node->left = insert(node->left,key);
    return node;
}

int main(){
    Node *root = NULL;
```

```
        int n,val;
    cout<<"Enter number of nodes to insert:";
    cin>>n;
    cout<<"Enter the values:";
    for(int i=0; i<n;i++){
        cin>>val;
        root=insert(root,val);
    }
    cout<<"Inorder Traversal of BST:";
    inorder(root);
    cout<<endl;
    cout<<"Preorder Traversal of BST:";
    preorder(root);
    cout<<endl;
    cout<<"Postorder Traversal of BST:";
    postOrder(root);
    cout<<endl;
    return 0;
}
```

OUTPUT:

```
C:\Users\ROHIT\OneDrive\De  X  +  v
Enter number of nodes to insert:5
Enter the values:5
4
3
2
1
Inorder Traversal of BST:1 2 3 4 5
Preorder Traversal of BST:5 4 3 2 1
Postorder Traversal of BST:1 2 3 4 5

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Process exited after 7.288 seconds with return value 0
Press any key to continue . . . |
```